PHYSICS 5054 QUESTION BANK - AUDIT
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Written 2026-11-13 11:28 UTC by tools/bank/breports.py.

RESULT: ALL AUTOMATED CHECKS PASSED; VISUAL REVIEW RECORDED

1. FILES
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  physics_5054_past_paper_questions_with_ms.pdf   79,617,744 bytes   4,358 pages
  physics_5054_question_bank_tracker.txt   1,487,174 bytes
  physics_5054_question_bank_tracker.pdf   804,027 bytes
  physics_5054_question_bank_exclusions.txt   49,937 bytes
  physics_5054_question_bank_exclusions.pdf   23,579 bytes
  physics_5054_question_bank_audit.txt   this file
  physics_5054_question_bank_audit.pdf   PDF copy of this file
  sha256 of the bank PDF: 1c68400f822ce0873885ffea32f486b91d9d4816c447649be947827f4e9aea97
  built 2026-11-13 10:44 UTC by tools/bank/bbuild.py; checked 2026-11-13 10:52 UTC by tools/bank/bverify.py

2. WHAT IS IN THE BANK
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  records (one question in one subsection)   2,709   Paper 1: 1,723   Paper 2: 986
  original questions   2,274   Paper 1: 1,723   Paper 2: 551
  Paper 2 questions by number of subsections   1 subsection: 263, 2 subsections: 177, 3 subsections: 82, 4
    subsections: 22, 5 subsections: 7
  pages   4,358   front 2, subsection title pages 27, pages of records
    4,329
  pictures placed   4,538   distinct pictures: 3,228 (a question printed under
    several subsections uses the same pictures)
  bookmarks   2,741   chapter and front matter 5, subsections 27, records 2,709
  page orientation   portrait 3,923, landscape 435 (landscape mark schemes from 2017 on)
  papers   156 question papers with their mark schemes: 78 of Paper 1, 78 of
    Paper 2, May/June 2004 to May/June 2026
  EITHER/OR questions   5 questions, 6 records, both alternatives and both schemes shown;
    the marks in the header are those of one alternative:
      4.2.2|w21|P2|V2|Q5   4 of 5 marks (printed total)
      4.2.3|w09|P2|V1|Q11   2 of 15 marks (printed marks above the OR alternative)
      4.2.4|s21|P2|V2|Q8   3 of 6 marks (printed total)
      4.2.4|s19|P2|V1|Q8   2 of 6 marks (printed total)
      4.2.4|s10|P2|V1|Q7   3 of 5 marks (printed marks above the OR alternative)
      4.2.4|w09|P2|V1|Q11   7 of 15 marks (printed marks above the OR alternative)
  pictures   16 grey levels, 180 dpi or more, shown at original size; white on
    black (grey values of the rendered source page inverted)
  Paper 1 answers   1,723   typeset as text below the question ("Answer: C"), from
    the key row of the mark scheme; no picture of the key
  order   subsection; then newest first: year 2026 down to 2004,
    October/November before May/June of the same year; variant 1 before 2; Paper 1 before Paper 2; question number
                                                  first record 1.1|s26|P1|V1|Q1, last record 4.2.4|s04|P2|V1|Q6

3. RECONCILIATION

[PASS] R1 the records of the bank are exactly the question/subsection pairs that the verified findings give
       2,709 pairs expected from the findings, 2,709 records in the bank, 2,709 in both
[PASS] R2 the entry list used for the build equals a fresh list made from the findings, in the same order
       2,709 entries, order identical
[PASS] R3 questions of the 156 papers: in the bank + left out = all questions
       Paper 1: 1,723 + 1,397 = 3,120; Paper 2: 551 + 257 = 808; checked for each of the 156 papers
[PASS] R4 no question is both in the bank and in the list of exclusions
       0 questions in both lists
[PASS] R5 papers without a mark scheme: every question is listed and left out
       10 papers, 200 + 55 = 255 questions
[PASS] R6 all questions of all Paper 1 and Paper 2 question papers in the folder
       2,274 in the bank + 1,654 not on a selected subsection + 255 without a mark scheme = 4,183
[PASS] R7 pages of the bank: front pages + divider pages + pages of the records = all pages
       2 + 27 + 4,329 = 4,358; the pages of consecutive records follow each other without a gap
[PASS] R8 files of the folder physics-5054: used + excluded + other kinds = all
       311 + 12 + 471 = 794
[PASS] R9 consolidation: several tracker rows of one question in one subsection give one record
       1,985 in-scope tracker rows of Paper 2 give 986 records of 551 questions; 1,723 Paper 1 questions give one
         record each
[PASS] R11 all papers used are complete in the paper-by-paper verification; Paper 1: the recorded answer equals the
       published key letter
       78 Paper 1 papers: 3,120 of 3,120 answers equal the key, 0 differ or are open; 78 Paper 2 papers: 3,673
         sub-questions matched to their scheme entries
[PASS] R10 the results of the checks belong to the files on disk
       PDF hash equals the hash recorded by bverify.py; geometry hash equals the hash recorded by bverify.py and
         bcheck_geom.py

  Questions by source   questions   in the bank   left out   reason for leaving out
  Paper 1, papers with a mark scheme   3,120   1,723   1,397   not on a selected subsection
  Paper 2, papers with a mark scheme   808   551   257   not on a selected subsection
  Paper 1, papers without a mark scheme   200   0   200   no mark scheme (98 on a selected
    subsection)
  Paper 2, papers without a mark scheme   55   0   55   no mark scheme (35 on a selected

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    subsection, 59 question/subsection pairs)
  all                                        4,183         2,274       1,909

4. AUTOMATED CHECKS OF THE CROP GEOMETRY (tools/bank/bcheck_geom.py, against the source PDFs)
-----------------------------------------------------------------------------------------
  run 2026-11-13 10:33 UTC; 156 papers
[PASS] V1 Paper 2: the mark allocations printed inside the crops of a question add up to the marks of that question
       808 questions, 0 differ
[PASS] V2 question papers: every text row and every answer line lies in exactly one crop
       48,856 rows, 0 outside or in two crops (509 ruled lines on the shared lined pages are not reproduced, see the
         exclusions file)
[PASS] V3 every question and every scheme starts inside the first crop of its own question
       3,928 question starts, 0 misplaced
[PASS] V4 Paper 2 schemes: every row of the used scheme pages lies in exactly one crop, or is general material listed
       in the exclusions file
       10,173 rows; 103 rows are in no crop: 77 after the scheme of the last question, 26 before the scheme of
         Question 1 (covers, general notes, legends of the mark codes; listed in the exclusions file, part D)
[PASS] V5 Paper 1: one crop per question; its text equals the question text of the verification
       3,120 questions, 0 differ
[PASS] V6 Paper 1: the key row of a question (the place its answer is read from) holds the question number and the
       key letter of that question, nothing else
       3,120 key rows, 0 differ
[PASS] V7 the pixel row of every cut line is free of ink (more than 8 dark pixels at 2 pixels per point), or the
       place was inspected
       10,628 cut lines, 2 with ink: s04_2 qp p. 11 at y=795.6; s25_22 qp p. 6 at y=796.5 (the two places described
         under V11)
[PASS] V8 Paper 2 schemes: no two neighbouring questions whose mark codes are off by opposite amounts (a row filed
       under the wrong question)
       0 such pairs
[PASS] V9 every question crop holds text (no crop of ruled lines only)
       0 crops without text
[PASS] V10 no ink is left outside the crops between page header and page footer
       2,610 question-paper pages and 468 scheme pages; 0 pages with unexplained ink; 19 shared lined pages not
         reproduced; 16 scheme pages with the general material of V4
[PASS] V11 no stroke crosses a cut line (8 pixels per point; excepted: cuts placed on a table rule, the rule of the
       examiner column), or the place was inspected
       10,628 cut lines, 2 crossed: s04_2 qp p. 11 at y=795.6: the mark '[9]', last line of Question 10, touches the
         page footer '[Turn over' (gap 0.12 pt, found at 8 pixels per point); the cut is at the lower end of the
         brackets and '[9]' is complete in the picture; s25_22 qp p. 6 at y=796.5: the corner marks of the page (page
         furniture, at x = 35-38 pt and 557-560 pt) stand beside the last answer line of Question 4; the cut is in
         the blank row below that line and passes through the corner marks only
  Notes of the geometry program:
    s04_2: QP page 11: last line of the question touches the page footer: bottom edge put at y=795.6, at a gap of
      0.12 pt found at 8 pixels per point
    s26_22: MS page 11: frame or rule ends directly above the page footer: bottom limit moved down to the blank row
      at y=549.75

5. AUTOMATED CHECKS OF THE FINISHED PDF (tools/bank/bverify.py: the PDF reopened from disk, every page rendered)
-------------------------------------------------------------------------------------------------------------
[PASS] A1 page count of the PDF equals the build manifest
       4358 pages
[PASS] A2 every entry is in the PDF and no record ID occurs twice
       2709 entries in page order, 2709 expected
[PASS] A3 order: subsection; newest first (year descending, October/November before May/June); variant, paper,
       question number
       strictly increasing sort key; first record 1.1|s26|P1|V1|Q1, last record 4.2.4|s04|P2|V1|Q6
[PASS] A4 one record per original question and subsection
       2274 original questions, 2709 records
[PASS] A5 pages of an entry are adjacent; question pictures first, complete scheme directly after, in source order
       (Paper 1: the answer line directly after the question)
       0 entries with a problem
[PASS] A6 bookmarks: one per chapter, subsection and entry, each pointing to the first page of its target
       5 top-level, 27 subsection and 2709 question bookmarks
[PASS] A7 file size below 100 MB
       79.6 MB (79617744 bytes)
[PASS] B0 every page of the PDF renders without a MuPDF error or warning
       4358 pages rendered; warnings: none
[PASS] B1 every page is white on black: no page is light (every page rendered; a page counts as light if more than 40
       % of it is bright)
       0 light pages of 4358; the brightest page is 13 % bright (page 978)
[PASS] B2 no blank page and no empty picture
       0 findings
[PASS] B3a every picture is on its page at the planned position (picture boxes read back from the PDF)
       4538 placements, 0 not found
[PASS] B3b the stored pixels of every distinct picture are identical to the inverted fresh rendering of its region of
       the source PDF
       3228 distinct pictures, 0 differ
[PASS] B3c every page, rendered on the pixel grid of its pictures, shows exactly the stored pictures (nothing over
       them, none displaced)
       4538 placements compared pixel by pixel, 0 differ
[PASS] B4 nothing printed outside the pictures, the answer lines and the bars; the rest of every page is black
       0 pages with stray ink
[PASS] B5 pictures lie inside the page area, below the header and above the footer, without overlap; orientation as
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planned; no page rotation
0 findings
[PASS] B6 the cut lines (outermost pixel rows of every picture) are free of ink; cuts on a table rule excepted
6453 cut lines checked, 3 on a ruled line by design; 6 with ink, of which 6 reviewed and accepted (listed in the audit), 0 not reviewed
[PASS] B7 every picture is shown at its original size (99.9 % or more) and at 180 dpi or more
4538 placements; scales {0.999: 50, 1.0: 3486, 1.001: 1002}; lowest resolution 180 dpi
[PASS] C1 header, source lines, record ID and page number of every page agree with the entry
0 pages with a missing text
[PASS] C2 marks of the question named in the header of every Paper 2 record = marks printed on the question paper inside its pictures
986 records of 551 questions: 980 records without alternatives (sum of all printed '[n]' and the printed '[Total: n]'), 6 records of EITHER/OR questions (marks of one alternative); 0 differ
[PASS] C3 Paper 1: the answer typeset on the page = the key letter in the row of that question in the mark scheme (read again from the source PDF) = the key of the verified findings
1723 answer lines of 1723 Paper 1 records; 0 differ; letters {'A': 407, 'B': 454, 'C': 431, 'D': 431}

Cut lines with ink that were inspected by eye and accepted (B6):

- bank page 528, bottom edge, 5054_s05_qp_2.pdf source page 2: the mark '[1]' of the last answer line ends directly above the page footer; the brackets are complete and nothing crosses the cut line
- bank page 866, bottom edge, 5054_s04_qp_2.pdf source page 11: the mark '[9]', last line of Question 10, touches the footer '[Turn over' (gap 0.12 pt); the cut is at the lower end of the brackets, '[9]' is complete; the last pixel row may carry a faint grey trace of the top of the footer text
- bank page 2594, bottom edge, 5054_s25_qp_22.pdf source page 6: the corner marks of the page (page furniture) stand beside the last answer line of Question 4; the cut is in the blank row below that line and passes through the two corner marks only; nothing of the question is cut
- bank page 2885, bottom edge, 5054_s26_ms_22.pdf source page 11: the frame of the note 'bod' below the table ends 0.5 pt above the page footer; the frame is complete
- bank page 4103, bottom edge, 5054_s23_qp_21.pdf source page 5: '[Total: 6]' ends directly above the footer '[Turn over'; the brackets are complete and nothing crosses the cut line
- bank page 4106, bottom edge, 5054_s23_qp_22.pdf source page 11: '[Total: 9]' ends directly above the footer '[Turn over'; the brackets are complete and nothing crosses the cut line

6. RECORDS PER SUBSECTION

subsection		Paper 1	Paper 2	records	bank pages
1.1	Physical quantities and measurement techniques	156	26	182	3-229
1.2	Motion	100	76	176	230-537
1.3	Mass and weight	73	39	112	538-717
1.4	Density	57	35	92	718-867
1.5.1	Balanced and unbalanced forces	85	62	147	868-1118
1.5.2	Friction	66	45	111	1119-1309
1.5.3	Elastic deformation	62	22	84	1310-1432
1.5.4	Circular motion	52	15	67	1433-1525
1.5.5	Turning effect of forces	59	37	96	1526-1679
1.5.6	Centre of gravity	41	14	55	1680-1759
1.6	Momentum	8	11	19	1760-1798
1.7.1	Energy	68	83	151	1799-2085
1.7.2	Work	38	30	68	2086-2199
1.7.3	Energy resources	27	17	44	2200-2272
1.7.4	Efficiency	41	29	70	2273-2389
1.7.5	Power	60	48	108	2390-2573
3.1	General properties of waves	89	53	142	2574-2798
3.2.1	Reflection of light	55	9	64	2799-2877
3.2.2	Refraction of light	76	37	113	2878-3052
3.2.3	Thin lenses	71	22	93	3053-3179
3.2.4	Dispersion of light	29	13	42	3180-3243
3.3	Electromagnetic spectrum	53	46	99	3244-3410
3.4	Sound	115	38	153	3411-3623
4.2.1	Electrical charge	90	29	119	3624-3783
4.2.2	Electrical current	35	40	75	3784-3932
4.2.3	Electromotive force and potential difference	44	29	73	3933-4062
4.2.4	Resistance	73	81	154	4063-4358
all		1723	986	2709	

7. RECORDS PER PAPER

paper	question paper	mark scheme	questions	in the bank	records	left out
s26_11	5054_s26_qp_11.pdf	5054_s26_ms_11.pdf	40	26	26	14
s26_21	5054_s26_qp_21.pdf	5054_s26_ms_21.pdf	9	7	10	2
s26_12	5054_s26_qp_12.pdf	5054_s26_ms_12.pdf	40	24	24	16
s26_22	5054_s26_qp_22.pdf	5054_s26_ms_22.pdf	9	5	11	4
w25_11	5054_w25_qp_11.pdf	5054_w25_ms_11.pdf	40	24	24	16
w25_21	5054_w25_qp_21.pdf	5054_w25_ms_21.pdf	9	6	10	3
w25_12	5054_w25_qp_12.pdf	5054_w25_ms_12.pdf	40	25	25	15
w25_22	5054_w25_qp_22.pdf	5054_w25_ms_22.pdf	9	5	12	4
s25_11	5054_s25_qp_11.pdf	5054_s25_ms_11.pdf	40	20	20	20
s25_21	5054_s25_qp_21.pdf	5054_s25_ms_21.pdf	9	5	12	4
s25_12	5054_s25_qp_12.pdf	5054_s25_ms_12.pdf	40	22	22	18
s25_22	5054_s25_qp_22.pdf	5054_s25_ms_22.pdf	9	6	11	3
w24_11	5054_w24_qp_11.pdf	5054_w24_ms_11.pdf	40	26	26	14
w24_21	5054_w24_qp_21.pdf	5054_w24_ms_21.pdf	9	5	10	4
w24_12	5054_w24_qp_12.pdf	5054_w24_ms_12.pdf	40	22	22	18
w24_22	5054_w24_qp_22.pdf	5054_w24_ms_22.pdf	9	4	7	5

s24_11	5054_s24_qp_11.pdf	5054_s24_ms_11.pdf	40	21	21	19
s24_21	5054_s24_qp_21.pdf	5054_s24_ms_21.pdf	8	6	10	2
s24_12	5054_s24_qp_12.pdf	5054_s24_ms_12.pdf	40	25	25	15
s24_22	5054_s24_qp_22.pdf	5054_s24_ms_22.pdf	9	6	10	3
w23_11	5054_w23_qp_11.pdf	5054_w23_ms_11.pdf	40	19	19	21
w23_21	5054_w23_qp_21.pdf	5054_w23_ms_21.pdf	9	6	12	3
w23_12	5054_w23_qp_12.pdf	5054_w23_ms_12.pdf	40	24	24	16
w23_22	5054_w23_qp_22.pdf	5054_w23_ms_22.pdf	10	7	11	3
s23_11	5054_s23_qp_11.pdf	5054_s23_ms_11.pdf	40	21	21	19
s23_21	5054_s23_qp_21.pdf	5054_s23_ms_21.pdf	11	6	8	5
s23_12	5054_s23_qp_12.pdf	5054_s23_ms_12.pdf	40	21	21	19
s23_22	5054_s23_qp_22.pdf	5054_s23_ms_22.pdf	10	5	10	5
w22_11	5054_w22_qp_11.pdf	5054_w22_ms_11.pdf	40	21	21	19
w22_21	5054_w22_qp_21.pdf	5054_w22_ms_21.pdf	9	7	13	2
w22_12	5054_w22_qp_12.pdf	5054_w22_ms_12.pdf	40	27	27	13
w22_22	5054_w22_qp_22.pdf	5054_w22_ms_22.pdf	9	6	9	3
s22_11	5054_s22_qp_11.pdf	5054_s22_ms_11.pdf	40	25	25	15
s22_21	5054_s22_qp_21.pdf	5054_s22_ms_21.pdf	9	6	13	3
s22_12	5054_s22_qp_12.pdf	5054_s22_ms_12.pdf	40	25	25	15
s22_22	5054_s22_qp_22.pdf	5054_s22_ms_22.pdf	9	6	12	3
w21_11	5054_w21_qp_11.pdf	5054_w21_ms_11.pdf	40	24	24	16
w21_21	5054_w21_qp_21.pdf	5054_w21_ms_21.pdf	10	7	13	3
w21_12	5054_w21_qp_12.pdf	5054_w21_ms_12.pdf	40	23	23	17
w21_22	5054_w21_qp_22.pdf	5054_w21_ms_22.pdf	9	7	14	2
s21_11	5054_s21_qp_11.pdf	5054_s21_ms_11.pdf	40	22	22	18
s21_21	5054_s21_qp_21.pdf	5054_s21_ms_21.pdf	10	6	13	4
s21_12	5054_s21_qp_12.pdf	5054_s21_ms_12.pdf	40	24	24	16
s21_22	5054_s21_qp_22.pdf	5054_s21_ms_22.pdf	11	9	15	2
w20_11	5054_w20_qp_11.pdf	5054_w20_ms_11.pdf	40	26	26	14
w20_21	5054_w20_qp_21.pdf	5054_w20_ms_21.pdf	10	7	11	3
w20_12	5054_w20_qp_12.pdf	5054_w20_ms_12.pdf	40	24	24	16
w20_22	5054_w20_qp_22.pdf	5054_w20_ms_22.pdf	9	5	12	4
s20_11	5054_s20_qp_11.pdf	5054_s20_ms_11.pdf	40	25	25	15
s20_21	5054_s20_qp_21.pdf	5054_s20_ms_21.pdf	10	7	11	3
s20_12	5054_s20_qp_12.pdf	5054_s20_ms_12.pdf	40	27	27	13
s20_22	5054_s20_qp_22.pdf	5054_s20_ms_22.pdf	10	7	12	3
w19_11	5054_w19_qp_11.pdf	5054_w19_ms_11.pdf	40	21	21	19
w19_21	5054_w19_qp_21.pdf	5054_w19_ms_21.pdf	10	5	10	5
w19_12	5054_w19_qp_12.pdf	5054_w19_ms_12.pdf	40	22	22	18
w19_22	5054_w19_qp_22.pdf	5054_w19_ms_22.pdf	10	7	13	3
s19_11	5054_s19_qp_11.pdf	5054_s19_ms_11.pdf	40	22	22	18
s19_21	5054_s19_qp_21.pdf	5054_s19_ms_21.pdf	11	8	16	3
s19_12	5054_s19_qp_12.pdf	5054_s19_ms_12.pdf	40	28	28	12
s19_22	5054_s19_qp_22.pdf	5054_s19_ms_22.pdf	11	7	11	4
w18_11	5054_w18_qp_11.pdf	5054_w18_ms_11.pdf	40	23	23	17
w18_21	5054_w18_qp_21.pdf	5054_w18_ms_21.pdf	10	7	15	3
w18_12	5054_w18_qp_12.pdf	5054_w18_ms_12.pdf	40	20	20	20
w18_22	5054_w18_qp_22.pdf	5054_w18_ms_22.pdf	10	8	16	2
s18_11	5054_s18_qp_11.pdf	5054_s18_ms_11.pdf	40	20	20	20
s18_21	5054_s18_qp_21.pdf	5054_s18_ms_21.pdf	11	8	13	3
s18_12	5054_s18_qp_12.pdf	5054_s18_ms_12.pdf	40	23	23	17
s18_22	5054_s18_qp_22.pdf	5054_s18_ms_22.pdf	11	7	14	4
w17_11	5054_w17_qp_11.pdf	5054_w17_ms_11.pdf	40	22	22	18
w17_21	5054_w17_qp_21.pdf	5054_w17_ms_21.pdf	11	5	14	6
w17_12	5054_w17_qp_12.pdf	5054_w17_ms_12.pdf	40	21	21	19
w17_22	5054_w17_qp_22.pdf	5054_w17_ms_22.pdf	11	10	20	1
s17_11	5054_s17_qp_11.pdf	5054_s17_ms_11.pdf	40	26	26	14
s17_21	5054_s17_qp_21.pdf	5054_s17_ms_21.pdf	11	7	13	4
s17_12	5054_s17_qp_12.pdf	5054_s17_ms_12.pdf	40	23	23	17
s17_22	5054_s17_qp_22.pdf	5054_s17_ms_22.pdf	11	8	13	3
w16_11	5054_w16_qp_11.pdf	5054_w16_ms_11.pdf	40	21	21	19
w16_21	5054_w16_qp_21.pdf	5054_w16_ms_21.pdf	10	7	13	3
w16_12	5054_w16_qp_12.pdf	5054_w16_ms_12.pdf	40	24	24	16
w16_22	5054_w16_qp_22.pdf	5054_w16_ms_22.pdf	10	6	15	4
s16_11	5054_s16_qp_11.pdf	5054_s16_ms_11.pdf	40	27	27	13
s16_21	5054_s16_qp_21.pdf	5054_s16_ms_21.pdf	12	9	15	3
s16_12	5054_s16_qp_12.pdf	5054_s16_ms_12.pdf	40	20	20	20
s16_22	5054_s16_qp_22.pdf	5054_s16_ms_22.pdf	11	8	11	3
w15_11	5054_w15_qp_11.pdf	5054_w15_ms_11.pdf	40	20	20	20
w15_21	5054_w15_qp_21.pdf	5054_w15_ms_21.pdf	11	8	15	3
w15_12	5054_w15_qp_12.pdf	5054_w15_ms_12.pdf	40	24	24	16
w15_22	5054_w15_qp_22.pdf	5054_w15_ms_22.pdf	11	7	13	4
s15_11	5054_s15_qp_11.pdf	5054_s15_ms_11.pdf	40	23	23	17
s15_21	5054_s15_qp_21.pdf	5054_s15_ms_21.pdf	11	7	12	4
s15_12	5054_s15_qp_12.pdf	5054_s15_ms_12.pdf	40	20	20	20
s15_22	5054_s15_qp_22.pdf	5054_s15_ms_22.pdf	11	7	14	4
w14_11	5054_w14_qp_11.pdf	5054_w14_ms_11.pdf	40	19	19	21
w14_21	5054_w14_qp_21.pdf	5054_w14_ms_21.pdf	11	7	16	4
w14_12	5054_w14_qp_12.pdf	5054_w14_ms_12.pdf	40	18	18	22
w14_22	5054_w14_qp_22.pdf	5054_w14_ms_22.pdf	11	7	12	4
s14_11	5054_s14_qp_11.pdf	5054_s14_ms_11.pdf	40	23	23	17
s14_21	5054_s14_qp_21.pdf	5054_s14_ms_21.pdf	11	8	15	3
s14_12	5054_s14_qp_12.pdf	5054_s14_ms_12.pdf	40	21	21	19

s14_22	5054_s14_qp_22.pdf	5054_s14_ms_22.pdf	11	8	13	3
w13_11	5054_w13_qp_11.pdf	5054_w13_ms_11.pdf	40	21	21	19
w13_21	5054_w13_qp_21.pdf	5054_w13_ms_21.pdf	11	6	13	5
w13_12	5054_w13_qp_12.pdf	5054_w13_ms_12.pdf	40	19	19	21
w13_22	5054_w13_qp_22.pdf	5054_w13_ms_22.pdf	11	8	18	3
s13_11	5054_s13_qp_11.pdf	5054_s13_ms_11.pdf	40	24	24	16
s13_21	5054_s13_qp_21.pdf	5054_s13_ms_21.pdf	11	5	8	6
s13_12	5054_s13_qp_12.pdf	5054_s13_ms_12.pdf	40	24	24	16
s13_22	5054_s13_qp_22.pdf	5054_s13_ms_22.pdf	11	9	17	2
w12_11	5054_w12_qp_11.pdf	5054_w12_ms_11.pdf	40	20	20	20
w12_21	5054_w12_qp_21.pdf	5054_w12_ms_21.pdf	11	10	16	1
w12_12	5054_w12_qp_12.pdf	5054_w12_ms_12.pdf	40	20	20	20
w12_22	5054_w12_qp_22.pdf	5054_w12_ms_22.pdf	11	8	12	3
s12_11	5054_s12_qp_11.pdf	5054_s12_ms_11.pdf	40	21	21	19
s12_21	5054_s12_qp_21.pdf	5054_s12_ms_21.pdf	11	9	13	2
s12_12	5054_s12_qp_12.pdf	5054_s12_ms_12.pdf	40	21	21	19
s12_22	5054_s12_qp_22.pdf	5054_s12_ms_22.pdf	10	7	11	3
w11_11	5054_w11_qp_11.pdf	5054_w11_ms_11.pdf	40	21	21	19
w11_21	5054_w11_qp_21.pdf	5054_w11_ms_21.pdf	11	8	14	3
w11_12	5054_w11_qp_12.pdf	5054_w11_ms_12.pdf	40	21	21	19
w11_22	5054_w11_qp_22.pdf	5054_w11_ms_22.pdf	11	8	15	3
s11_11	5054_s11_qp_11.pdf	5054_s11_ms_11.pdf	40	22	22	18
s11_21	5054_s11_qp_21.pdf	5054_s11_ms_21.pdf	11	8	11	3
s11_12	5054_s11_qp_12.pdf	5054_s11_ms_12.pdf	40	22	22	18
s11_22	5054_s11_qp_22.pdf	5054_s11_ms_22.pdf	11	8	12	3
w10_11	5054_w10_qp_11.pdf	5054_w10_ms_11.pdf	40	20	20	20
w10_21	5054_w10_qp_21.pdf	5054_w10_ms_21.pdf	10	9	14	1
w10_12	5054_w10_qp_12.pdf	5054_w10_ms_12.pdf	40	20	20	20
w10_22	5054_w10_qp_22.pdf	5054_w10_ms_22.pdf	11	8	16	3
s10_11	5054_s10_qp_11.pdf	5054_s10_ms_11.pdf	40	23	23	17
s10_21	5054_s10_qp_21.pdf	5054_s10_ms_21.pdf	11	9	14	2
s10_12	5054_s10_qp_12.pdf	5054_s10_ms_12.pdf	40	23	23	17
s10_22	5054_s10_qp_22.pdf	5054_s10_ms_22.pdf	11	8	14	3
w09_1	5054_w09_qp_1.pdf	5054_w09_ms_1.pdf	40	20	20	20
w09_2	5054_w09_qp_2.pdf	5054_w09_ms_2.pdf	11	8	13	3
s09_1	5054_s09_qp_1.pdf	5054_s09_ms_1.pdf	40	18	18	22
s09_2	5054_s09_qp_2.pdf	5054_s09_ms_2.pdf	11	6	11	5
w08_1	5054_w08_qp_1.pdf	5054_w08_ms_1.pdf	40	20	20	20
w08_2	5054_w08_qp_2.pdf	5054_w08_ms_2.pdf	11	8	15	3
s08_1	5054_s08_qp_1.pdf	5054_s08_ms_1.pdf	40	20	20	20
s08_2	5054_s08_qp_2.pdf	5054_s08_ms_2.pdf	11	7	11	4
w07_1	5054_w07_qp_1.pdf	5054_w07_ms_1.pdf	40	20	20	20
w07_2	5054_w07_qp_2.pdf	5054_w07_ms_2.pdf	11	7	11	4
s07_1	5054_s07_qp_1.pdf	5054_s07_ms_1.pdf	40	22	22	18
s07_2	5054_s07_qp_2.pdf	5054_s07_ms_2.pdf	11	8	13	3
w06_1	5054_w06_qp_1.pdf	5054_w06_ms_1.pdf	40	17	17	23
w06_2	5054_w06_qp_2.pdf	5054_w06_ms_2.pdf	11	8	12	3
s06_1	5054_s06_qp_1.pdf	5054_s06_ms_1.pdf	40	21	21	19
s06_2	5054_s06_qp_2.pdf	5054_s06_ms_2.pdf	11	7	9	4
w05_1	5054_w05_qp_1.pdf	5054_w05_ms_1.pdf	40	21	21	19
w05_2	5054_w05_qp_2.pdf	5054_w05_ms_2.pdf	11	9	16	2
s05_1	5054_s05_qp_1.pdf	5054_s05_ms_1.pdf	40	18	18	22
s05_2	5054_s05_qp_2.pdf	5054_s05_ms_2.pdf	11	7	14	4
w04_1	5054_w04_qp_1.pdf	5054_w04_ms_1.pdf	40	20	20	20
w04_2	5054_w04_qp_2.pdf	5054_w04_ms_2.pdf	10	7	8	3
s04_1	5054_s04_qp_1.pdf	5054_s04_ms.pdf	40	20	20	20
s04_2	5054_s04_qp_2.pdf	5054_s04_ms.pdf	11	6	11	5
all			3928	2274	2709	1654

8. VISUAL REVIEW

Record of the inspection by eye, written by the AI agent that built the bank. It describes the file physics_5054_past_paper_questions_with_ms.pdf with sha256 1c68400f822ce087 (first 16 digits), 4,358 pages. "Viewed" means: the page was rendered from the PDF on disk and the image was looked at.

8.1 What was viewed

* Contact sheets of ALL 4,358 pages of the file described here (41 sheets of small page images written by bverify.py), read at half size. Thin white strokes fade when a dark sheet is reduced, so the copies that were read had been passed through a maximum filter and brightened before the reduction. Every page is black with light print; no page is white or partly white; no picture was left black on white; no page is blank; the landscape mark schemes stand upright; the front pages and the 27 title pages are in place.
* 59 pages at reading size (rendered at 97 dpi, landscape pages at 130 dpi):
  front page, contents, first title page (3): 1-3
  Paper 1, question with its answer line (13): 4, 228, 558, 1011, 1307, 2119, 2157, 2687, 2707, 2792, 2861, 3065, 3754
  Paper 2, question pages (31): 233-234, 309, 369, 373, 472, 571, 710, 1185, 2358, 2645, 2790, 3067, 3292, 3362, 3409, 3680, 3809-3810, 3830, 4047-4048, 4067, 4138-4139, 4169-4170, 4310, 4320-4321, 4357
  Paper 2, mark schemes on portrait pages (6): 3410, 3828, 4049, 4240, 4311, 4322
  Paper 2, mark schemes on landscape pages (6): from source pages that carry a page rotation of 90 degrees: 303, 2836, 3811, 4140, 4171; from a landscape source page without rotation: 235

  Chosen for: the first record (4) and the last (4357); the oldest papers (2004: 228, 1307, 3409-3410, 4357) and the newest (2026: 4, 233-235, 4067, with the pale watermark of the download site); the Paper 1 page on which the answer line stands lowest (2687); every page of the six records of the five EITHER/OR questions (3809-3811,

4047-4049, 4138-4140, 4169-4171, 4310-4311, 4320-4322); a scheme in small print at the end of a record of five pages, the longest there is (3828); schemes that begin with the heading "SECTION B" (2836, 3410); photographs and dark or shaded artwork (309, 369, 373, 558, 710, 1011, 2645, 2790, 3065, 3292, 3362, 3680), among them the record pages that check B1 measures as the lightest (373, 3292, 3362, 3680, 710, 2790; pages 978, 1473 and 3485 show the pictures of 373 and 3292 under other subsections); a random sample of 16 pages (seed 5054: 303, 472, 571, 1185, 1307, 2119, 2157, 2358, 2707, 2792, 2861, 3067, 3754, 3830, 4067, 4240).
On all of them the question or scheme is complete, upright and white on black, the print is sharp, the header names the right paper and question, and the scheme follows its question. On the Paper 1 pages the line "Answer: X" stands below the question, under a bar that names the mark scheme file, its page and the question; on 2687 it stands clear of the page footer. The EITHER/OR records show both alternatives and both schemes, and their header says so.

* Paper 1 answers against the printed key. For 12 further records chosen at random (seed 5054; pages 184, 215, 626, 712, 1280, 1587, 1821, 2446, 3062, 3401, 3492, 3755) the header and the answer line of the bank were put beside the answer key of the source mark scheme, rendered as printed, with the row of the question marked. The letter is the same in all 12. (Check C3 makes this comparison for all 1,723 records by reading the words of the row from the source PDF.)
* The bookmarks: the first 34, the last three of subsection 1.1, the first three and the last three of subsection 1.2 and the last three of the file were read. They run from May/June 2026 down to May/June 2004, October/November before May/June of the same year.
* The PDF copies of the reports, white on black: tracker pages 1, 2 and 150, exclusions pages 1 and 8, audit pages 1 and 7.

8.2 Pictures and headers are those of the file inspected before

* The black-on-white file of commit f410326 (sha256 1c4c44d6e6440b49, 4,358 pages) was inspected in more detail: its 41 contact sheets were read, 83 pages were viewed at full size (of 20 of them the upper part) and 4 at 62 % size, and the six cut lines that carry ink (check B6) and the two cut lines that a stroke touches (V7, V11) were viewed enlarged beside their source pages. The record of that inspection is part 8 of the audit file of that commit.
* tools/bank/bneg.py read both files back from disk and compared them:
  - each of the 3,218 distinct pictures of the file described here is, pixel for pixel, the negative of a picture of that file;
  - each of the 2,709 records holds, page by page and from top to bottom, the pictures it held there. Only the last picture of each of the 1,723 Paper 1 records is gone: the row of the answer key, which is text now;
  - the text that the bank prints on the pages of the 986 Paper 2 records (header, source bars, footer) is word for word the same, apart from the number of the record and of the page. On the pages of the 1,723 Paper 1 records the label "QUESTION and ANSWER" stands for "QUESTION and MARK SCHEME" and the line "Answer: X" is new; nothing else differs.
* So no crop and no header changed with the colours and the order, and what was corrected after the earlier inspections is still as corrected: the marks and the remark of the EITHER/OR questions, the names of the parts in the header, the heading "SECTION B" kept with Question 9, three pictures with hatching, two pictures that were cut at the page footer, header lines no longer shortened. The cut lines of B6, V7 and V11 were not viewed again for this file.

8.3 Observations that are not defects

* Photographs and filled or shaded drawings are negatives. Large light areas are artwork that is black in print: 373, 978 and 1473 (astronaut in space, 5054_w17_qp_22 p. 13), 558 (5054_s24_qp_12 p. 3), 3362 (5054_s13_qp_22 p. 6), 3292 and 3485 (X-ray image of a hand, 5054_s20_qp_21 p. 17: the bones, light in print, are dark here). The man in the lift (309, 5054_w21_qp_21 p. 4) is a black figure in print and a white one here.
* The tones of a shaded drawing are reversed, and the wording of a question is not. Seen on 4067 (5054_s26_qp_21 p. 13, viewed beside the source page): the question says that the inside surface of a lamp "darkens"; the darkened part of the drawing is the lighter part here. Part 9 names the pages where the inversion shows most.
* The watermark of the download site in the papers of 2026, pale grey in print, is a dark grey here (4, 233-235, 4067).
* Page furniture inside pictures (described with counts in the exclusions file, part D2): the heading of a section above its first question (369, 373, 1185, 3409; in a scheme 2836, 3410); the page number in a Paper 2 paper of 2004 (4357); the column "For Examiner's Use" (3362, 4310, 4357); the vertical notice "DO NOT WRITE IN THIS MARGIN", which is cut where a picture ends (233, 234, 3067, 4067).

8.4 Defects found while the colours were changed, and corrected

* A test build of 60 records failed check B4 on 66 pages: the black background ended exactly on the page edge and the rendered page kept light pixels along its border. The background is now drawn 20 pt beyond the page on all sides. B4 passes on all 4,358 pages of the file described here.
* The PDF copies of the audit and of the exclusions were first set, white on black, in the regular cut of their typewriter font. Its thin strokes are dim on black. Both cuts were viewed side by side, and the two PDFs are now set in the bold cut.
* The exclusions file said that the notice "DO NOT WRITE IN THIS MARGIN" is in the papers of 2025 and 2026. Page 3067 shows it in a paper of October/November 2024. The file now names the papers that carry the notice, found by searching the text of all Paper 2 question papers.
* The inspection by eye of the bank found no defect in its pages.

8.5 Not inspected by eye

* 4,299 of the 4,358 pages were seen only as small images on the contact sheets (of 12 of them also the header and the answer line). At that size one can see that a page is complete, upright and white on black, but not read it. That every picture shows exactly its region of the source page, that every line of every question and scheme lies in a picture, and that every answer letter is the letter of the key, rests on the automated checks of parts 4 and 5.
* Of the report PDFs, only the pages named in 8.1.
* Whether each question is filed under the best subsection was not reviewed again in this inspection.

9. HOW THE BANK WAS MADE, AND LIMITS

* Source: the PDF files in the folder physics-5054 of the repository. Every picture of the bank is rendered directly from them
  by tools/bank/bbuild.py. No page of an earlier bank is used.
* The files of the repository are stored with inverted colours: each page content is wrapped in a block that paints the page
  white and inverts it with a blend mode. The wrapper also leaves white as the fill colour, so text that the original draws in
  the default colour is invisible in the stored copy (282 pages of 31 of the files used, mostly papers of 2004-2009).
  The bank removes the wrapper from the page content before rendering. That restores the page as it was printed, with the
  text that is invisible in the stored copies.
* Colours: the bank is white on black, as requested on 13 November 2026. The restored page is rendered black on white,
  reduced to 16 grey levels and then inverted, so that print is white and paper is black. Check B3b compares every picture
  with exactly this rendering, and check B1 finds no light page. The stored inverted copies are not used for it.
  Everything inside a picture is inverted, photographs and shaded drawings included: they appear as negatives. A part that
  a question calls black or dark is white or light in its picture, and the other way round.
  Where it shows most (measured, not judged): on 19 pages more than 4 % of the page is light, which means large black areas
  in the print (photographs, silhouettes, filled drawings):
    369, 373, 508, 710, 969, 978, 1101, 1473, 1934, 2076, 2149, 2328, 2384, 2565, 2790, 3292, 3362, 3485, 3680
  and in 37 of the 3,228 pictures 5 % or more of the pixels are mid-grey (grey-shaded artwork), on 61 pages:
    280, 283, 369, 417, 453, 471, 486, 490, 511, 558, 811, 860, 903, 969, 1011, 1030, 1071, 1229, 1270, 1324, 1647, 1671,
    1754, 1934, 2040, 2149, 2203, 2275, 2328, 2368, 2391, 2559, 2590, 2645, 2646, 2650, 2672, 2683, 2713, 2715, 2831, 3028,
    3065, 3093, 3111, 3116, 3127, 3135, 3160, 3197, 3198, 3387, 3426, 3526, 3548, 3550, 3616, 3680, 3890, 3899, 4269
  Example: page 4067 (May/June 2026, Paper 2 variant 1, Question 6(b)). The question says that the inside surface of a lamp
  "darkens" where metal atoms condense; in the bank the darkened part of the drawing is the lighter part.
* Paper 1 answers are text, as requested on 13 November 2026: "Answer: C" below the question, and above it a line with the
  name of the mark scheme file and the page. The letter is the key recorded in the verified findings. It is compared with
  the row of that question in the answer key when the bank is built, and again by check C3. There is no picture of the key.
* Order: newest first, as requested on 13 November 2026. Inside a subsection: year from 2026 down to 2004, October/November
  before May/June of the same year, then variant, paper and question number.
* The checks that are new with these changes were tried with planted faults. On a build of 60 records, a picture left
  black on white is reported by B1, B3b and B6, and a wrong answer letter by C1 and C3. On pages made for the purpose,
  check F12 of bfinal.py reports a white page, a white border and a page without print, and check F13 reports black or
  dark grey text on black and black text on white.
* The PDF copies of the tracker, of the exclusions and of this audit are white on black as well (checks F12 and F13
  render every page of them). The .txt copies are plain text and have no colours.
  Check F12 failed when it was first run, on all 16 pages of the exclusions and audit PDFs: rendered at half size, the
  thin strokes of their typewriter font were too dim to count as print. The check now renders at 108 dpi, and the two
  PDFs are set in the bold cut of that font, which is easier to read on black.
* Landscape mark schemes (2017 on) carry a page rotation or are landscape pages. The rotation is removed with the display
  transformation of the page before any coordinate is measured; these schemes are shown on landscape pages, upright.
* A question is cut out over the full page width from the blank band above its first line to the blank band below its last
  line; a question that runs over several source pages gives one picture per page. The cut between two questions is placed
  from the ink of the rendered page, not from text recognition. Nothing inside a picture is moved, removed or compressed.
* Which subsection a question belongs to comes from the paper-by-paper verification of this project (every question read with
  its mark scheme by the AI agent that built the bank, plus mechanical checks). It is a judgement, not an official
  Cambridge classification. Questions written for earlier syllabuses are filed under the 2026-2028 subsection they fit.
* The automated checks compare pixels, positions, page text and counts. They cannot judge whether a question is filed under
  the best subsection. The visual review (part 8) covered contact sheets of all pages and full-size samples, not every page
  at full size.
* Third-party marks in the source files (a watermark and footer of the download site in the 2026 files) are part of the source
  pages; the footer is outside the pictures, the pale watermark is visible inside them.